

ISS 2026 AI Hackathon

Track 3: AI for Climate and Environmental Health

Data Acquisition Guide

Overview

Welcome to Track 3: AI for Climate and Environmental Health.

This challenge requires participants to develop artificial intelligence solutions that leverage climate, environmental, geospatial, and demographic data to address public health challenges in Nigeria.

Unlike Tracks 1 and 2, no single dataset is provided. Participants are expected to obtain publicly available datasets from the official sources listed below, integrate them into a single analytical dataset, engineer relevant features, and develop an AI solution.

Challenge Objective

Develop an AI solution that predicts or explains climate-sensitive public health risks in Nigeria.

Possible applications include:

- Malaria risk prediction
- Cholera hotspot prediction
- Flood vulnerability assessment
- Heat stress prediction
- Environmental health surveillance
- Climate-sensitive disease forecasting

Required Datasets

1. **NASA POWER Climate Data:** <https://power.larc.nasa.gov/data-access-viewer/>

Download the following variables

- Average Temperature (T2M)
- Maximum Temperature (T2M_MAX)
- Minimum Temperature (T2M_MIN)
- Relative Humidity (RH2M)
- Precipitation (PRECTOTCORR)
- Wind Speed (WS2M)
- Solar Radiation (ALLSKY_SFC_SW_DWN)

Recommended Temporal Resolution

- Monthly

Geographic Coverage

- Nigeria

2. **CHIRPS Rainfall Dataset:** <https://www.chc.ucsb.edu/data/chirps>

Download:

- Monthly Rainfall
- Daily Rainfall

Coverage:

- Nigeria

3. **WorldPop Population Dataset:** <https://www.worldpop.org/>

Download:

- Nigeria Population Density

Recommended Variable:

- Population Density

4. **OpenStreetMap:** <https://www.openstreetmap.org/>

Useful Layers

- Roads
- Rivers
- Settlements
- Buildings
- Health Facilities

These data may be used to derive variables such as:

- Distance to nearest river
- Distance to nearest health facility
- Road accessibility

5. **SRTM Digital Elevation Model:** <https://earthexplorer.usgs.gov/>

Download:

- Nigeria Elevation Data

Recommended Variables

- Elevation
- Slope

Optional Dataset

6. MODIS Vegetation Index (NDVI): <https://search.earthdata.nasa.gov/>

Recommended Variable

- NDVI

Vegetation information may improve models for vector-borne diseases such as malaria.

Data Integration

- Participants are expected to merge all downloaded datasets into a unified analytical dataset.

Recommended merge variables include:

- Latitude
- Longitude
- Month
- Year

The resulting dataset should contain both environmental and demographic variables suitable for machine learning.

Expected Dataset Structure

Variables include:

- Latitude
- Longitude
- State
- Local Government Area (optional)
- Month

- Year
- Average Temperature
- Maximum Temperature
- Minimum Temperature
- Relative Humidity
- Rainfall
- Wind Speed
- Solar Radiation
- Elevation
- Slope
- Population Density
- NDVI (optional)

Participants may engineer additional variables where appropriate.

Suggested Prediction Tasks

Participants may choose one of the following:

- **Malaria Risk Prediction:** Predict areas with increased malaria transmission risk using climate and environmental variables.
- **Flood Vulnerability Prediction:** Predict communities at risk of flooding.
- **Heat Stress Prediction:** Predict areas with elevated heat stress risk.
- **Cholera Risk Prediction:** Predict areas vulnerable to cholera outbreaks.

Participants may propose additional climate-health applications, provided they clearly justify the public health relevance.

Deliverables

Each team must submit:

1. Source Code (GitHub Repository)
2. Technical Report
3. Presentation Slides

4. Five-minute Demonstration Video

Please note that the technical report should clearly describe:

1. Data sources
2. Data preprocessing
3. Feature engineering
4. Model development
5. Model evaluation
6. Limitations
7. Public health implications

Evaluation Criteria

S/N	Criterion	Weight
1	Data Acquisition and Preprocessing	20%
2	Data Integration and Feature Engineering	20%
3	AI Model Performance	30%
4	Explainability and Interpretation	15%
5	Innovation and Public Health Impact	15%

Rules

1. Only publicly available datasets should be used.
2. Teams must clearly cite all data sources.
3. Additional open datasets may be incorporated provided licensing permits their use.
4. All preprocessing and feature engineering steps must be documented.
5. Teams should ensure that their solution is reproducible.

Support

Participants are encouraged to consult the official documentation for each dataset and may raise technical questions through the hackathon communication channels.

We wish all participants success in developing innovative AI solutions that advance climate resilience and public health in Nigeria.